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Integrated Stochastic Scheduling of Renewable-Based Virtual Power Plants in Sequential Markets Considering Activation-Aware Reserve Bidding and N-1 Reliability

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Abstract:	Participation of renewable-based Virtual Power Plants (VPPs) in German sequential electricity markets requires sophisticated bidding strategies that account for complex auction structures, uncertain reserve activations, and operational reliability. This paper proposes AlocaBid, a stochastic Mixed-Integer Linear Programming (MILP) framework that integrates day-ahead energy trading, secondary (SRL) and tertiary (MRL) reserve markets, and N-1 reliability constraints within a seven-stage rolling-horizon approach aligned with actual market timelines. Key innovations include activation-aware reserve bidding using three probabilistic scenarios derived from four years of historical 50Hz TSO data, and dispatch groups that decouple asset allocation from product-specific commitments, enabling flexible portfolio reconfiguration without asset-level restructuring. The framework is implemented in Python with the Gurobi solver and validated through four comprehensive case studies. Results demonstrate that AlocaBid outperforms rule-based Straight-Forward Allocation (SFA) by 33% to 201% across diverse market conditions. Activation revenues contribute 60–70% of reserve market profitability, challenging conventional strategies that prioritize capacity payments. The N-1 constraint reduces available flexibility by only 5–8% but improves reserve delivery success from 90.7% to 98.8%, representing a favorable trade-off between economic performance and operational security. Comparative analysis confirms MILP superiority over metaheuristic algorithms (GA: 18.6%, PSO: 17.8%), while scalability analysis demonstrates near-linear execution time for portfolios of up to 100 assets. The AlocaBid tool bridges the gap between theoretical optimization and practical VPP operations, offering a decision-support system aligned with German market timelines and reliability requirements.
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Dear Professor

Editor-in-Chief
International Journal of Electrical Power & Energy Systems

We are pleased to submit our original manuscript entitled "***Integrated Stochastic Scheduling of Renewable-Based Virtual Power Plants in Sequential Markets Considering Activation-Aware Reserve Bidding and N-1 Reliability***" for consideration in the *International Journal of Electrical Power & Energy Systems*.

This work presents **AlocaBid**, a novel stochastic MILP framework for optimal scheduling of renewable-based VPPs in German sequential electricity markets. The framework integrates three key innovations:

(i) activation-aware reserve bidding using probabilistic scenarios derived from four years of 50Hz TSO data,

(ii) a seven-stage rolling-horizon approach aligned with weekly (SRL), daily (MRL), and hourly day-ahead market timelines, and

(iii) N-1 reliability constraints to ensure reserve deliverability under single-asset contingencies—a feature rarely addressed in VPP literature.

Our case studies across four market scenarios demonstrate that AlocaBid consistently outperforms rule-based allocation, achieving profit improvements of 33% to 201%, with activation revenues contributing 60–70% of reserve profitability. The N-1 constraint enhances reserve delivery success from 90.7% to 98.8% at a minimal 5–8% flexibility reduction. Benchmark comparisons confirm MILP superiority over metaheuristic methods, while scalability analysis validates practical applicability for portfolios of up to 100 assets.

Your kind consideration and attention is highly appreciated. There are some other articles that have been published in *IJEPES* which have the same scope as this paper. These articles are used as the references in this paper:

[3] Mohamed Saad Suliman, Hooman Farzaneh, Data-driven modeling of the aggregator-based price-maker virtual power plant (VPP) in the day-ahead wholesale electricity markets; evidence from the Japan Electric power Exchange (JEPX) market, International *Journal of Electrical Power & Energy Systems*, Volume 164, **2025**, 110433, https://doi.org/10.1016/j.ijepes.2024.110433.

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[26] Peng Sun, Mao Yang, Teng Yun, Zhe Chen, Dynamic aggregation willingness prediction and operation decision of multi-energy virtual power plant based on graph attention and multi-agent federated reinforcement learning, *International Journal of Electrical Power & Energy Systems*, Volume 181, **2026**, 112101, https://doi.org/10.1016/j.ijepes.2026.112101.

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This manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All authors have approved the submission and declare no conflicts of interest.

We believe this work aligns closely with *IJEPES*'s scope, particularly in power system operation, stochastic optimization, renewable integration, and system reliability. We would be honored to have it considered for publication and look forward to your feedback.

Yours sincerely,

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Highlights

- Proposes a rolling-horizon stochastic MILP for VPP bidding in German sequential markets.
- Integrates activation-aware reserve bidding with probabilistic activation scenarios.
- Introduces dispatch groups for flexible portfolio reconfiguration.
- Incorporates N-1 reliability constraints to ensure secure reserve delivery.

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Integrated Stochastic Scheduling of Renewable-Based Virtual Power Plants in Sequential Markets Considering Activation-Aware Reserve Bidding and N-1 Reliability

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ABSTRACT Participation of renewable-based Virtual Power Plants (VPPs) in German sequential electricity markets requires sophisticated bidding strategies that account for complex auction structures, uncertain reserve activations, and operational reliability. This paper proposes AlocaBid, a stochastic Mixed-Integer Linear Programming (MILP) framework that integrates day-ahead energy trading, secondary (SRL) and tertiary (MRL) reserve markets, and N-1 reliability constraints within a seven-stage rolling-horizon approach aligned with actual market timelines. Key innovations include activation-aware reserve bidding using three probabilistic scenarios derived from four years of historical 50Hz TSO data, and dispatch groups that decouple asset allocation from product-specific commitments, enabling flexible portfolio reconfiguration without asset-level restructuring. The framework is implemented in Python with the Gurobi solver and validated through four comprehensive case studies. Results demonstrate that AlocaBid outperforms rule-based Straight-Forward Allocation (SFA) by 33% to 201% across diverse market conditions. Activation revenues contribute 60–70% of reserve market profitability, challenging conventional strategies that prioritize capacity payments. The N-1 constraint reduces available flexibility by only 5–8% but improves reserve delivery success from 90.7% to 98.8%, representing a favorable trade-off between economic performance and operational security. Comparative analysis confirms MILP superiority over metaheuristic algorithms (GA: 18.6%, PSO: 17.8%), while scalability analysis demonstrates near-linear execution time for portfolios of up to 100 assets. The AlocaBid tool bridges the gap between theoretical optimization and practical VPP operations, offering a decision-support system aligned with German market timelines and reliability requirements.

INDEX TERMS Virtual Power Plant (VPP), Secondary Reserve (SRL), Tertiary Reserve (MRL), Stochastic MILP, Dispatch Group, N-1 Reliability, Reserve Activation, Day-Ahead Market, AlocaBid, Portfolio Optimization.

Nomenclature

Abbreviations

DA	Day-Ahead Market
DG	Dispatch Group
DR	Demand Response
FiT	Feed-in Tariff
MILP	Mixed Integer Linear Programming
MRL	Tertiary Reserve (Minute reserve)
NDA	Non-Disclosure Agreement
PV	Photovoltaic
SFA	Straight-Forward Allocation
SRL	Secondary Reserve (Sekundärregelleistung)
TSO	Transmission System Operator
VPP	Virtual Power Plant
WAC	Weighted Average Cost

***Set* Symbol**

ω	Scenario index for reserve activations
t	Time index (hours)
a	Asset index
d	Dispatch group (DG) index
f	Product index (SRL or MRL)
r	Rank index in merit order
i	Stage index (days of the week)

Parameters

λ_t	Electricity price at time t ($/MWh)
c_c	Production cost of non-subsidized assets ($/MWh)
$\delta_{t,a}$	Power sold by asset a at time t (MW)
FiT_a	Feed-in tariff for subsidized asset a ($/MWh)
$\Psi_{T,a}$	Penalty for non-delivery of contracted power ($/MWh)
$P_{S,a}$	Contracted power amount for asset a (MWh/day)

$P^{+}_{asset,t,d}$	Maximum allowed power output of asset a at time t (MW)
$P^{-}_{asset,t,d}$	Minimum required power output of asset a at time t (MW)
$P_{commit,a}$	Technical minimum commitment power of asset a (MW)
$P_{max,a}$	Maximum capacity of asset a (MW)
$F_{prequal,a}$	Prequalified flexibility volume of asset a (MW)
$F_{min,f}$	Minimum bid volume for product f (MW)
$F_{min,commit,f}$	Minimum commitment volume for product f (MW)
$F_{prequal,f}$	Prequalified capacity for product f (MW)
M	Big-M constant
μ_f	Price coefficient for reserve type f ($/MW)
$\pi_{f,\omega}$	Probability of activation scenario ω for product f
$A_{f,r,\omega}$	Activation hours for rank r in scenario $\omega$for product f
WAC_f	Weighted average cost for product f ($/MWh)
$\hat{k}$	Penalty cost for DG transitions ($/MW)
$Ramp\ Up$	Maximum upward ramp rate of asset a (MW/h)
$Ramp\ Down$	Maximum downward ramp rate of asset a (MW/h)

Variables

Π_i	Spot market profit in stage i ($)
RM	Reserve market profit ($)
SM	Spot market profit ($)
F_f	Bid volume for product f (MW)
Z_f	Expected activation profit for product f ($)
$F_{a,t,d}$	Flexibility offered by asset a in DG d at time t (MW)
$F_{dg,d}$	Total flexibility of dispatch group d (MW)
$y_{t,a}$	Binary commitment of asset a at time t
$y^{SRL}_{d,a}$	Binary indicating assignment of asset a to SRL in DG d
$y^{MRL}_{d,a}$	Binary indicating assignment of asset a to MRL in DG d
$y_{dg,d}$	Binary activation of dispatch group d
$y_{dg,d,a}$	Binary commitment of asset a to DG d
x_f	Binary indicator of active bid for product f
v_r	Binary indicator of selected rank r
$g^{SRL}_{t,d,a}$	SRL flexibility committed by asset a in DG d at time t (MW)
$g^{MRL}_{t,d,a}$	MRL flexibility committed by asset a in DG d at time t (MW)
$z^{SRL}_{i,a}$	Binary indicating SRL DG change for asset a in stage i
$z^{MRL}_{i,a}$	Binary indicating MRL DG change for asset a in stage i
$\omega_{t,d,a}$	Binary identifying largest asset in DG d at time t
$soldF$	Sold flexibility in previous stages (MW)

I. INTRODUCTION

A. Background

Unlike vertically integrated markets, Germany's decentralized market design, with its four TSOs and distinct reserve product timelines, creates opportunities and complexities for VPP operators that demand sophisticated scheduling tools. Over the last decade, the growing emphasis

on reducing carbon emissions has significantly increased the deployment of renewable energy resources in modern power systems. This shift, while environmentally beneficial, introduces significant operational challenges due to the intermittent nature of wind and solar generation. To manage the increasing number of distributed energy resources, virtual power plants (VPPs) have gradually evolved into an effective coordination mechanism capable of combining geographically dispersed assets into a single operational entity. The German electricity market provides an ideal context for VPP development because of its advanced framework for renewable integration and well-established reserve market structures. In this market, VPPs can trade energy in the DA market and offer flexibility in secondary (SRL) and tertiary (MRL) reserve markets, creating multiple revenue streams from the same portfolio of assets.

B. Literature Review

B.1 Uncertainty Management in VPP Scheduling

Virtual power plants have emerged as a transformative concept in modern power system operations, enabling the aggregation and coordinated management of distributed energy resources. The evolving regulatory and operational frameworks for energy communities, as comprehensively reviewed by Bashi et al. [1], underscore the continued viability of VPPs as a key solution for integrating renewable energy sources while maintaining grid stability. Subsequent research has expanded this concept to address various operational challenges, including market participation, uncertainty management, and resource coordination [2,

3]. Recent studies have proposed advanced optimization models that improve upon earlier deterministic approaches. Kim and Choi [4] developed a sample robust optimal bidding model for VPPs, while Shahkoomahalli et al. [5] proposed a risk-based electrical-thermal scheduling framework incorporating downside risk constraints for large-scale VPPs in energy and reserve markets. These initial formulations, while valuable, largely ignored the inherent uncertainties associated with renewable generation and market prices. To address uncertainty, researchers have employed robust and stochastic optimization techniques, as exemplified by the improved light robust strategy proposed by Wu et al. [6] and the comprehensive review of optimization methods by Nafkha-Tayari et al. [7].

The stochastic nature of renewable generation and electricity prices has motivated extensive research on uncertainty management in VPP operations. Recent contributions have explored various methodological frameworks, including robust optimization [8, 9], distributionally robust optimization [10], and hybrid stochastic-robust approaches [11]. Safari and Rahbarimagham [12] proposed a flexible-reliable AC security-constrained unit commitment framework, whereas Bharti et al. [13] demonstrated the effectiveness of ANN-MILP hybrid techniques for EV-PV-BESS integration. The comparative analysis of uncertainty management approaches by Lima et al. [14] reveals that stochastic programming generally provides less conservative solutions than robust optimization, though robust methods offer stronger guarantees against worst-case outcomes. This trade-off has motivated the development of hybrid approaches that balance economic optimality with operational security [15]. The integration of conditional value at risk (CVaR) has become increasingly popular for managing financial risk in VPP operations, as evidenced by the works of Zhang et al. [16] and Li et al. [17].

B.2 Reserve Market Participation & Activation Modeling

While early VPP studies focused on energy market participation, recent research has increasingly addressed reserve market integration. The recognition of VPPs as flexibility providers has led to comprehensive models that account for secondary (SRL) and tertiary (MRL) reserve provision [18, 19]. Naval and Yusta [20] provided a comprehensive review of VPP models in electricity markets, identifying reserve market participation as a key research gap. The economic value of flexibility provision is closely tied to activation revenues, which remain underexplored in the existing literature. Most studies focus on capacity payments, overlooking the potentially higher revenues from actual reserve activations. Fernández-Muñoz et al. [21] developed optimization models for day-ahead energy and reserve self-scheduling, whereas Bagchi et al. [22] examined the adequacy of generating systems incorporating storage-integrated VPPs.

Several recent studies have pointed out that reserve activation payments can play a much larger economic role than previously assumed, particularly in balancing markets with frequent activations. While capacity payments have received significant attention, the revenue from actual activations remains underexplored. Some researchers have proposed sequential bidding approaches to reflect the temporal structure of electricity markets, yet comprehensive models that simultaneously handle multiple reserve products with distinct time horizons are scarce.

The increasing deployment of battery energy storage systems in VPPs has prompted research on degradation-aware operations. Studies by Abdulla et al. [23] and Hauer et al. [24] demonstrated the importance of incorporating battery lifecycle models in VPP scheduling. Alimardani and Narimani [25] proposed new energy storage system configurations to extend Li-ion battery lifetime in household applications. More recently, Ma et al. [26] developed multi-time scale scheduling strategies for VPPs considering source-load uncertainty and demand response. What sets our approach apart is not only the use of stochastic programming, but also the combination of activation-aware bidding, reliability constraints, and a market-aligned sequential structure—elements that existing models treat separately, if at all. Demand response programs have become integral to VPP operations, enabling load shaping and improved flexibility provision. Wang et al. [27] and Lazović and Đurišić [28] significantly explored various DR mechanisms, including price-based and incentive-based approaches. The coordinated scheduling of heterogeneous generation assets, including renewable generation and storage systems, has been shown to significantly enhance VPP economic performance in German electricity markets through optimized day-ahead scheduling that accounts for compensation structures and cost characteristics [29]. Recent advances in artificial intelligence and machine learning have opened new possibilities for VPP optimization. Liu et al. [30] reviewed the applications of reinforcement learning in VPP decision-making, while Ruan et al. [31] provided a comprehensive analysis of data-driven energy management approaches. The integration of deep learning for forecasting [32, 33] and optimization [34] represents an active research frontier.

B.3 Reliability, Dispatch Groups, and sequential market structure

Recent developments in VPP operation have increasingly focused on the coordinated participation of aggregated DERs in multiple electricity markets while maintaining adequate reliability standards. As VPP portfolios grow in size and diversity, ensuring secure operation under uncertain market conditions has become a critical challenge. Several studies have therefore emphasized the importance of incorporating reliability-oriented constraints into market scheduling models. In particular, reliability-based reserve scheduling frameworks have demonstrated that adequate reserve allocation is essential for maintaining operational security when renewable generation and flexible demand resources are exposed to significant uncertainty. These

studies highlight that reserve commitments should maximize economic returns and guarantee the ability of the VPP to withstand unexpected asset outages or forecasting errors.

The concept of dispatch groups, which decouple asset allocation from product-specific commitments, has been introduced in industry practice but lacks rigorous academic treatment. Similarly, the N-1 reliability constraint, crucial for ensuring secure delivery of reserve commitments, is often overlooked in VPP optimization models. In parallel, the concept of coordinated dispatch among aggregated resources has emerged as a key component of modern VPP operation. Rather than scheduling each resource independently, recent studies advocate the use of aggregated dispatch regions and coordinated control schemes that allow multiple assets to act as a unified market participant. This approach improves operational flexibility, facilitates reserve provision, and enhances the ability of the VPP to satisfy market requirements. The notion of dispatch groups and feasible operating regions has proven particularly effective in representing the collective behavior of heterogeneous assets while preserving the physical limitations of individual resources.

The interaction between energy and reserve markets has also received considerable attention in recent years. Unlike simplified formulations that assume simultaneous market clearing, many real-world electricity markets follow a sequential auction structure in which energy commitments are determined before reserve procurement and balancing actions. Studies investigating sequential market participation have demonstrated that decisions made in earlier market stages directly influence the flexibility available for subsequent reserve and balancing services. Thus, integrated scheduling approaches that explicitly represent sequential market clearing processes can capture important operational dependencies and produce more economically efficient bidding strategies. To the best of our knowledge, no existing study simultaneously integrates activation-aware reserve bidding, N-1 reliability constraints, dispatch-group allocation, and sequential market timing within a single stochastic MILP framework. Most models focus on one or two of these aspects, missing the synergies—and practical constraints—that matter in real German reserve markets. This gap is the starting point for our AlocaBid framework, which brings all these pieces together in a way that is both theoretically sound and operationally viable. Table 1 compares key features of recent VPP models including uncertainty handling, N-1 constraints, dispatch groups, and sequential optimization. [51]

Table 1. Comprehensive Literature Comparison of VPP Scheduling Models

Ref.	Year	Approach	Markets	Uncertainty	N-1	DG	Seq.
[35]	2021	Review	E	-	-	-	-
[36]	2024	Review	E	-	-	-	-
[37]	2023	Robust	E	Price	✗	✗	✗
[38]	2022	Stochastic	E	Wind, Price	✗	✗	✗
[39]	2016	Hybrid	E	Price, Load	✗	✗	✗
[40]	2020	Stochastic	E	Price, Wind	✗	✗	✗
[41]	2022	Stochastic	E+SRL+MRL	Price, Wind, Solar	✗	✗	✗
[42]	2019	Stochastic	E+SRL	Generation	✗	✗	✗
[43]	2025	Multi-stage	E	Source, Load	✗	✗	✗
[44]	2023	Stochastic	E+SRL	Wind	✗	✗	✓
[45]	2024	MILP	E	None	✗	✗	✗
[46]	2023	RL	E+SRL	Multi-source	✗	✗	✗
[47]	2024	Data-driven	E	Gen., Price	✗	✗	✗
[48]	2023	Stochastic	E+SRL+MRL	Price, Wind, Solar	✗	✗	✓
[49]	2024	Robust	E+SRL	Gen., Load	✗	✗	✗
[50]	2024	Stochastic	E+SRL+MRL	Price, Wind, Solar	✓	✓	✓
This Paper	2026	Stochastic	E+SRL+MRL	Price, Wind, Solar	✓	✓	✓

By introducing dispatch groups and the N-1 reliability constraint, we address these limitations. In conventional reserve scheduling models, each asset is directly linked to a specific reserve product. Such a formulation creates rigid allocation structures and limits the ability of the VPP operator to reconfigure the portfolio in response to changing market conditions. In contrast, the proposed Dispatch Group concept separates asset commitment from product nomination. Assets are first aggregated into flexibility pools and subsequently assigned to reserve products according to market opportunities.

From an operational perspective, this structure allows portfolio managers to react more effectively to changing market signals without repeatedly restructuring individual asset commitments. In direct allocation approaches, the number of possible assignments is approximately M^N. By introducing DGs, assets are grouped into intermediate flexibility clusters, allowing multiple reserve products to share the same flexibility resource. Hence, the model can exploit additional combinations that are infeasible under direct allocation.

As a result, the optimization process gains access to a broader set of feasible reserve combinations, which can lead to economically superior bidding decisions.

Unlike existing stochastic VPP scheduling approaches that primarily focus on energy-market participation, the proposed framework jointly integrates activation-aware reserve bidding, dispatch-group flexibility aggregation, and N−1 reliability

management within a sequential market structure. This combination enables a closer representation of practical German reserve-market operations.

Table 2. Research Gap Analysis

Gap	Existing Studies	Proposed Work
Activation revenue	Partial	Full stochastic modeling
N-1 reliability	Rare	Integrated
Dispatch groups	Not addressed	Proposed
Sequential bidding	Limited	Weekly framework
Practical implementation	Rare	AlocaBid

Table 2 presents a comparative analysis of key research gaps addressed by the proposed framework. Existing studies usually focus on either market participation or operational reliability. A framework capable of simultaneously addressing sequential bidding, reserve activation uncertainty, portfolio flexibility management, and reliability requirements has not yet been reported. The proposed framework addresses these gaps directly. Finally, the AlocaBid tool bridges the gap between theoretical optimization and practical implementation. This approach greatly expands the operator's options and makes it easier to adapt the portfolio to changing market opportunities. Table 1 and 2 together paint a clear picture: while many studies have tackled individual pieces—uncertainty, reserve bidding, or reliability—none have woven them into a single, practical framework. Activation revenues are often treated as an afterthought, N-1 constraints are rare outside transmission planning, and dispatch groups remain a purely industrial concept without academic formalization. Our work directly addresses these gaps by embedding activation-aware bidding, N-1 reliability, and dispatch-group flexibility into a sequential MILP structure that mirrors the actual German market timeline. The AlocaBid tool, described later, turns this theoretical integration into a usable operational system.

C. Main Contributions

The main contributions of this paper are as follows:

- A stage-based sequential MILP formulation adapted to capture the temporal structure of the DA, SRL (weekly), and MRL (daily) markets through a seven-stage horizon.
- Formalization of the dispatch group concept, extending industry practice into a rigorous optimization framework
- Application of N-1 reliability constraints to VPP scheduling, previously limited to transmission-level planning
- Stochastic activation modeling incorporating three activation scenarios derived from historical grid data
- The AlocaBid tool, implemented in Python with Gurobi, is designed for practical VPP operations
- A comprehensive case study comparing AlocaBid against straight-forward allocation across different market conditions

Comprehensive case study: Four scenarios comparing AlocaBid against straight-forward allocation (SFA) across different market conditions and portfolio compositions.

A novel activation-scenario generation methodology based on four years of empirical 50Hz TSO data, provides a realistic representation of reserve deployment patterns.

D. Paper Structure

The remainder of the article is organized as follows. Section II presents the mathematical formulation of the proposed model, including the stage-based sequential approach, objective function, and all constraints. Section III describes the technical-structural model implementation, including portfolio construction and the AlocaBid tool. Section IV presents the case studies and simulation results across four scenarios: base case, high SRL prices, high MRL prices, and big assets portfolio, followed by sensitivity analysis and benchmark comparisons. Section V discusses the results, limitations, and provides conclusions with suggestions for future research.

II. MODEL FORMULATION

A. Stage-Based Sequential Modeling

Products in the German electricity market have different time horizons. SRL products are traded weekly, MRL products have daily auctions, and DA products are traded hourly. To capture this structure, the model optimizes a full week divided into seven stages, where each day T corresponds to stage i. This approach provides two key advantages: (1) it aligns with actual auction timing, reflecting the sequential nature of bidding, and (2) it enables practical implementation in operational systems. The stage-based approach inherently creates a trade-off in portfolio management. While a large portfolio increases uncertainty in forecasting asset failures, it also enables the management of unexpected failures through backup reserves based on the N-1 criterion. Fig. 1 shows the weekly bidding timeline for the DA, SRL, and MRL markets with their respective time horizons.

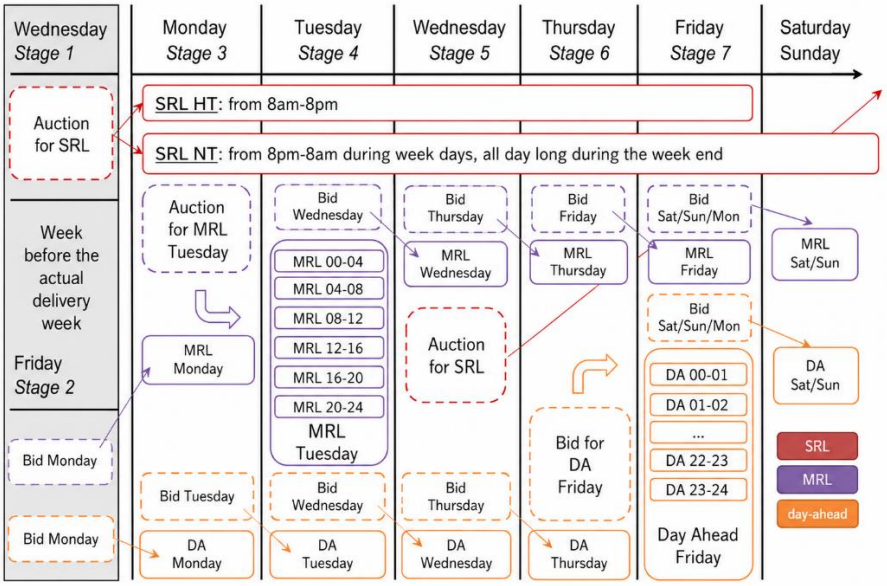


Fig. 1 illustrates the timeline of different stages and bidding schedules across DA, SRL, and MRL markets throughout the week.

B. Sequential vs. simultaneous optimization

In the operational version of the model, a sequential approach is adopted since operators run it daily. However, the theoretical formulation assumes all stages occur simultaneously, which simplifies the mathematical representation and allows focusing on cases where failure information is known in advance. In practice, the model is executed sequentially on a daily basis. At each stage *j*, the optimization is performed for the remaining horizon (from stage *j* to 7), while respecting the irrevocable commitments made in previous stages. This rolling-horizon approach, formalized in constraint (34), allows the VPP operator to adapt to newly available information, such as updated price forecasts or asset statuses, without violating the strategic decisions of the past.

For the purpose of mathematical exposition and to prove the concept, the optimization problem is formulated as a single-stage model that optimizes the entire week simultaneously. This formulation provides the theoretical foundation and allows us to focus on the structure of the constraints without the added complexity of sequential decision-making.

In practice, the model is executed daily in a rolling-horizon fashion. At each stage j, the optimization covers the remaining horizon (stages j to 7) while respecting irrevocable commitments from previous stages. Constraint (34) enforces this by deducting the cumulative sold flexibility from the total available capacity of each dispatch group, ensuring that past bids are never violated when new decisions are made.

C. Portfolio Characteristics

The VPP portfolio consists of subsidized and non-subsidized assets. Subsidized assets receive feed-in tariffs for their generation, whereas non-subsidized assets participate directly in the spot market. The model assumes that the portfolio includes generation assets only, without demand response loads.

Assumptions:

1. Scenario generation captures uncertainties (including DA prices) for better VPP modeling.
2. Two-stage "planning" and "operational" optimization for risk management.
3. Focus on profitability and optimal resource allocation according to the mathematical model.

Market assumptions:

- DA prices include stochastic scenarios for uncertainty modeling.
- Capacity prices for SRL and MRL are input parameters based on historical data from E.ON.
- VPP has a forecast of the energy merit order curve, with negligible impact of its bid on total supply.
- Assets have ramp-up/ramp-down rates and dynamic marginal costs (WAC).
- Demand response (DR) is incorporated as flexibility resource.

- Average monthly German price for subsidized assets is used for FiT calculation.

D. Objective Function

The objective is to maximize VPP profit with two revenue channels: the spot market and the reserve market.
Total Profit Function:

$$Tot = SM + RM \tag{1}$$

Spot Market Profit:

$$\Pi = \sum_i \Pi_i \tag{2}$$

Due to the stage-based approach, total spot market profit is the sum of daily spot market profits across stages.
Per-Stage Spot Market Profit:

$$\Pi_i = \sum_t \lambda_t \left(\sum_a \delta_{t,a} - c_c \sum_a \delta_{t,a} \right) + \sum_a FiT_a P_{S,a} - \sum_a \Psi_{T,a} \left(P_{S,a} - \frac{1}{24} \sum_t \delta_{t,a} \right) \tag{3}$$

Where:

- λ_t: Electricity price at time t
- c_c: Production cost of non-subsidized assets
- $\delta_{t,a}$: Power sold by assets
- FiT_a: Feed-in tariff for subsidized assets
- $\Psi_{T,a}$: Penalty for non-delivery of contracted power
- $P_{S,a}$: Contracted power amount

Reserve Market Profit:

$$RM = \sum_f \mu_f . F_f + \sum_f Z_f - \sum_t \sum_d \sum_a \hat{k} \left(g_{t,d,a}^{SRL} - g_{t,d,a}^{MRL} \right) \tag{4}$$

where $\mu_f$is the price coefficient for each reserve type, $F_f$is the capacity cost of reserve, $Z_f$is the expected activation profit, $g_{t,d,a}^{SRL}$is the SRL reserve amount for asset $a$in DG $d$at time t, $g_{t,d,a}^{MRL}$is the MRL reserve amount for asset $a$in DG $d$at time t, and $\hat{k}$is the penalty cost for DG transitions. Most existing models assign each asset directly to one reserve product, which limits flexibility. This penalty, while small in magnitude, serves two purposes: it stabilizes the schedule and reduces computational time by discouraging unnecessary solver branching. We solve this by introducing dispatch groups and N-1 reliability constraints. We added a small penalty on DG switches to discourage frequent reallocation, which improves stability and cuts solve time. This keeps the schedule stable and reduces computation time. The final term in (4) imposes a small penalty on transitions of assets between dispatch groups across consecutive stages. This penalty discourages frequent reallocation of assets, which could incur operational costs or complicate the real-time management of the VPP.
where $\lambda_f$is the price coefficient for each reserve type, $c_f$is the capacity cost of reserve, $\pi_f$is the expected activation profit, $r_{a,d,t}^{SRL}$is the SRL reserve amount for asset $a$in DG $d$at time t, $r_{a,d,t}^{MRL}$is the MRL reserve amount for asset $a$in DG $d$at time t, and $\rho$is the penalty cost for DG transitions. Most existing models assign each asset directly to a specific reserve product, which severely limits operational flexibility. We address this by introducing dispatch groups and the N-1 constraint. The final term in the objective function adds a small penalty on unnecessary changes between dispatch groups. This helps keep the schedule stable and reduces computational effort.

E. Activation Modeling

For activation revenue estimation, two elements are essential: (1) the bid energy price (λ), and (2) the number of activated hours per week. The bid price determines the rank in the merit order list. Bids with lower prices are activated more frequently. However, during large imbalances, even high-priced assets are needed and earn substantial activation revenue. Three Activation Scenarios:

1. High activations
2. Medium activations
3. Low activations

For each product, three activation curves are generated with different probabilities ($\pi_{f,\omega}$) depending on the product's nature. To capture the full economic potential of the VPP portfolio, the model incorporates demand response (DR) as an additional flexibility resource. DR programs enable load shaping through price-based and incentive-based mechanisms, allowing the VPP to reduce consumption during peak price periods and increase flexibility for reserve market participation. The DR revenue is modeled as a function of the load reduction capacity, the DR price signal, and the availability of flexible loads during peak hours. Additionally, the model accounts for price uncertainty in the day-ahead market by generating multiple price scenarios

based on historical price distributions. This stochastic treatment of DA prices ensures that the scheduling decisions remain robust to market fluctuations, a critical requirement for practical VPP operations where price forecasts are inherently uncertain.
Expected Activation Profit:

$$Z_f = F_f \cdot \max\left(\sum_{\omega} \pi_{f,\omega} \cdot A_{f,r,\omega}(\lambda_{f,r} - WAC_f)\right) \quad (5)$$

The rank that provides the highest profit determines the energy price to be placed in the bid.

F. Constraints

1. Asset Constraints

Maximum Output Constraint:

$$\sum_d \delta_{t,a} + F^+_{a,t,d} \le P^+_{asset,t,d} \forall t, a \quad (6)$$

Ensures asset power output does not exceed the maximum allowed value.

$$\delta_{t,a} - \delta_{t-1,a} \le Ramp\ Up. y_{t,a} \quad (6\text{-}1)$$

$$\delta_{t-1,a} - \delta_{t,a} \le Ramp\ Down. y_{t,a} \quad (6\text{-}2)$$

Minimum Output Constraint:

$$\sum_d \delta_{t,a} - F^-_{a,t,d} \ge P^-_{asset,t,d} \forall t, a \quad (7)$$

Ensures asset power output does not fall below the minimum required value.
Technical Minimum Level:

$$\delta_{t,a} \ge P_{commit,a} y_{t,a} \forall t, a \quad (8)$$

Ensures committed assets produce above a technical minimum level.
Production Commitment:

$$\sum_t \delta_{t,a} = 24 P_{S,a} \forall a \quad (9)$$

The VPP schedules each asset's production according to its contracted amount.
Commitment Bounds:

$$\delta_{t,a} \le P_{commit,a} y_{t,a} \forall t, a \quad (10)$$

$$\delta_{t,a} \ge P_{commit,a} y_{t,a} \forall t, a \quad (11)$$

Dispatch Group Commitment:

$$\sum_a y_{dg,d,a} = y_{dg,d} \forall d \quad (12)$$

Positive Flexibility Provision:

$$F^+_{a,t,d} \le P_{max,a}(1 - y_{dg,d,a}) \forall t, a, d \quad (13)$$

Allows uncommitted assets to provide positive flexibility.
Prequalification Constraint:

$$F_{dg,d} \le F_{prequal,a} \forall t, a, d \quad (14)$$

Ensures assets do not offer more flexibility than their prequalified amount.

2. Dispatch Group Constraints

Total DG Flexibility:

$$\sum_a F_{a,t,d} = F_{dg,d} \forall d \quad (15)$$

Total flexibility of a DG equals the sum of all assets committed to it.
DG Activation Condition:

$$y_{dg,d} \ge y_{dg,d,a} \forall d, a \quad (16)$$

If an asset is committed to a DG, the DG must be active.
DG Active Count:

$$y_{dg,d} \le \sum_a y_{dg,d,a} \forall d \quad (17)$$

Ensures DG cannot have more active assets than those committed.
Concurrent Reserve Commitment:

$$y^{SRL}_{d,a} + y^{MRL}_{d,a} \le 1 \forall d, a \quad (18)$$

Prevents assets from being simultaneously committed to SRL and MRL.
SRL DG Change Condition:

$$z_{i,a}^{SRL} \geq y_{i,a}^{SRL} - y_{i-1,a}^{SRL} \forall i, a \quad (19)$$

SRL Minimum Commitment:

$$\sum_{a} g_{t,d,a}^{SRL} \geq M y_{i,a}^{SRL} \forall i, t, a \quad (20)$$

MRL DG Change Condition:

$$z_{i,a}^{MRL} \geq y_{i,a}^{MRL} - y_{i-1,a}^{MRL} \forall i, a \quad (21)$$

MRL Minimum Commitment:

$$\sum_{d} g_{t,d,a}^{MRL} \geq M y_{i,a}^{MRL} \forall i, t, a \quad (22)$$

3. Product Constraints

Lower Bound:

$$F_f \geq F_{min,f} x_f \forall f \quad (23)$$

Upper Bound:

$$F_f \leq F_{prequal,f} x_f \forall f \quad (24)$$

Minimum Commitment:

$$F_f \geq F_{min,commit,f} x_f \forall f \quad (25)$$

DG Filling Constraints:

$$\sum_{d} F_{dg,d} \geq F_f x_f \forall f \quad (26)$$

$$\sum_{d} F_{dg,d} \leq M x_f \forall f \quad (27)$$

4. N-1 Reliability Constraints

Let $\mathcal{O}$be the set of N-1 contingency scenarios where the largest committed asset in the dispatch group $d$at time $t$is unavailable. For each scenario $o \in \mathcal{O}$:

$$F_{dg,d,t} - F_f \geq \max_{a \in \mathcal{D}_d} \left(P_{\max,a} \cdot (1 - y_{dg,d,a}) \right) \forall f, d, t \quad (28)$$

Constraint (28) enforces the N-1 security criterion by ensuring that the available dispatchable flexibility, after the outage of the largest committed asset within each dispatch group, remains sufficient to satisfy the contracted reserve capacity. This conservative reliability requirement guarantees reserve deliverability under any single-asset contingency and enhances the operational credibility of the VPP in ancillary service markets. In the present formulation, the N-1 criterion is evaluated only for the largest committed asset at each time interval, while correlated outages are beyond the scope of this study. Therefore, the remaining flexibility after the loss of the largest asset must be greater than or equal to the committed reserve bid, thereby maintaining secure reserve provision under single-component failure conditions. Stochastic Activation Constraints. For instance, if a dispatch group has committed 10 MW of SRL and contains a 5 MW asset, the N-1 constraint requires the remaining assets to provide at least 10 MW after removing that 5 MW asset—effectively requiring the group to have 15 MW of total flexibility. Overbooking ensures reliability at a 5-8% cost in available capacity, as shown in our case studies.

Linearization of Max Operator:

$$Z_f \leq \sum_{\omega} \pi_{f,\omega} \cdot A_{f,r,\omega} (\lambda_{f,r} - WAC_f) F_f + M(1 - v_r) \forall f, r \quad (29)$$

$$Z_f \geq \sum_{\omega} \pi_{f,\omega} \cdot A_{f,r,\omega} (\lambda_{f,r} - WAC_f) F_f - M(1 - v_r) \forall f, r \quad (30)$$

$$\sum_{r} v_r = 1 \quad (31)$$

These constraints replace the non-linear max operator, making the model solvable.

G. Operational Sequential Bidding

Since the objective function cannot be directly applied to operational use (asset failures cannot be predicted), a sequential approach is employed. At each stage, incoming products are optimized:

$$_j \text{ Maximize } R = \sum_{i \geq j} \Pi_i + \sum_{i \geq j} \sum_{f \in S_i} Z_f \, ; j \in \{1,2, \dots, 7\} \quad (33)$$

Resource Allocation Constraint:

$$F_{dg,d,t} - \sum_{f \in \mathcal{N}_{d,t}} sold\, F_f \cdot B_{t,f} \geq 0 \forall t, d, f; j \geq i \quad (34)$$

The constraint (34) guarantees that the sum of flexibility that is reserved for the dispatch group *d* at stage *t* is enough to meet the demands in all previous stages. In this constraint, soldF indicates the sum of flexibility that has been sold in the previous stages of the process. The coupling constraint is the heart of the rolling-horizon approach and guarantees that the flexibility cannot be over-sold, since some resources have already been reserved for prior needs. Guarantees that resources reserved for active bids in previous stages are sufficient. It is worth noting that the N-1 constraint guarantees reliability but implies a compromise between reliability and profitability. Namely, in the case of big assets, the N-1 constraint decreases the flexibility in 5-8% in comparison with the base case. This is the price of reliable reserve supply. The N-1 constraint presupposes the simultaneous outage of the largest asset in each dispatch group. This assumption can be considered conservative and overestimates the amount of backup flexibility. Alternative formulations could consider partial outages or different failure probabilities for different asset types.

$$M = \max\left(P_{max,a}\right) + \sum_{f} F_{prequal,f} \tag{35}$$

M (for constraints 20, 22, 27, 29). And $M_{profit} = 10^6$(a sufficiently large number in Euros) for constraints (30) and (31).

H. Scenario Generation Methodology

Historical activation hours for both SRL and MRL over a period of four years were obtained from the public 50Hz TSO data. Probability distributions were fitted for both products and directions (Positive/Negative) of the weekly activation hours. High, Medium and Low scenarios were defined to be the 85th, 50th and 15th percentiles, respectively. The respective probabilities have been set as $\pi = \{0.2, 0.6 \text{ and } 0.8\}$. This method captures the tail risks while keeping the computations computationally feasible.

A log-normal distribution was fitted to the weekly activation hours via maximum likelihood estimation. The 15th, 50th and 85th percentiles are the low, medium and high scenarios, respectively.

In order to investigate the robustness of the proposed model framework, an extensive sensitivity analysis on key market parameters, namely DR prices, reserve market prices and ramp penalty, has been performed. The results of sensitivity analysis (Fig. 28) show that VPP profit is not sensitive to small changes in these key market parameters, meaning that the optimized schedule is robust enough to such price changes. This is a good characteristic of the practical operation where the forecasts can never be completely accurate. Moreover, comparison with meta-heuristic optimization algorithms (GA and PSO) shows that the proposed MILP model always performs better than the two approaches (rule-based and heuristic), generating profit increases in the range of 33-201% as opposed to the increases of 17.8-18.6% attained via GA and PSO (Fig. 26). Convergence graph (Fig. 27) proves the computational efficiency of the proposed optimization method whereas scalability analysis (Fig. 29) shows linear increase in execution time with respect to number of securities. Profit distribution (Fig. 30) once again proves that reserve profit makes up to 60-70% of VPP profit where activation profits greatly exceed capacity payments.

I. Model Limitations

Several limitations of the proposed model should be acknowledged:

Several limitations remain:

1. Data validation: Dataset modified for NDA; public 50Hz TSO validation shows 5-8% difference—results should be interpreted cautiously.
2. Computational burden: Solution time is 2.3h (warm-start) to 6h for 100 assets; Benders decomposition recommended.
3. Activation scenarios: Based on historical data; extreme events not fully covered.
4. Reliability: Partial outages and correlated failures not modeled.

Recent studies have demonstrated the growing importance of uncertainty management, reliability enhancement, and risk-aware decision-making in virtual power plant (VPP) operation. Pérez-Díaz [52] investigated the optimal participation of VPPs in energy and reserve markets and showed that coordinated bidding strategies can significantly improve market revenues while maintaining operational flexibility. To address the challenges associated with renewable generation uncertainty, Nemati [53] developed a comprehensive framework for the optimal operation and bidding of renewable-based virtual power plants under uncertainty, demonstrating that stochastic optimization approaches can significantly enhance economic performance in electricity markets while accounting for the inherent variability of renewable generation. In a related work, Conejo et al. [54] established fundamental methodologies for optimization and decision-making under uncertainty in electricity markets, which have become a cornerstone for modern stochastic scheduling models.

In terms of reliability, Bagchi et al. [55] showed that including energy storage systems into VPP architecture can help to increase generation adequacy and improve system reliability under uncertain conditions. More recently, Safari and Rahbarimagham [51] suggested a security-constrained scheduling approach that takes into account both uncertainty in renewable energy supply and system reliability needs, stressing the necessity to include security measures into the formulation of market-driven optimization problem. This all together shows that proper functioning of VPP should be based on the trade-off between profitability, uncertainties and reliability of the system and serves as motivation for development of the proposed risk-considering optimization framework.

J. *Risk-averse Extension Based on Conditional Value at Risk (CVaR)*

While the optimization framework introduced in this paper is intended to maximize the expected profit of the virtual power plant (VPP), decision making process in electricity markets is not based only on expected values. There is always some level of uncertainty associated with the possible reserve request, changing market clearing prices, errors in forecasting of renewable energy and limited availability of distributed energy resources. This means that even such operating approaches that can provide high expected profit can lead to serious financial losses in certain circumstances.

Considering such risks, it is possible to introduce the risk-averse variant of the model using the Conditional Value at Risk (CVaR) criterion. The Conditional Value at Risk is an established risk measure applied in power system analysis and electricity markets since it estimates the average loss from the worst outcomes that happen above a certain confidence level. CVaR is different from standard risk measures since it offers a consistent convex risk measure.

The risk-adjusted objective function is formulated as follows:

$$max\ Z = \sum_{s \epsilon S} p_s\, \Pi_s - \lambda CVaR_\alpha \tag{35}$$

$$CVaR_\alpha = \eta + \frac{1}{1-\alpha} \sum_{s \epsilon S} p_s\, \xi_s \tag{36}$$

$$\xi_s \geq L_s - \eta \tag{37}$$

$$\xi_s \geq 0 \tag{38}$$

$$L_s = -\Pi_s \tag{39}$$

Where Z is the value of the risk-adjusted objective function. S refers to the uncertainty scenarios set. S indicates a certain scenario. P_S is the probability of the particular scenario (s). Π_S represents the profit earned in scenario (s). L_S is the monetary loss of scenario (s). 〖CVaR〗 _α is the Conditional Value at Risk at the confidence level of (α). α is the confidence level of 0.90 to 0.99. η is an additional variable that corresponds to the Value-at-Risk (VaR). ξ_S is a non-negative additional variable that captures the excess loss above the VaR level of scenario (s). λ is the risk aversion coefficient that controls the balance between profitability and risk levels.

The Conditional Value at Risk (CVaR) risk averse model has been formulated (see Appendix A). The numerical realization of the model will be done later, since the present research is devoted to maximizing the expected profitability under the N 1 reliability approach. The higher value of the (λ) leads to more conservative bidding approach, in which safety is preferred to profit. On the contrary, if (λ=0), then the model becomes risk neutral and concentrates on maximizing the expected profitability only.

The integration of CVaR into the suggested VPP schedule allows for the development of a balanced approach regarding the tradeoff between economic and financial aspects. Therefore, the resulting strategy is more robust to poor market outcomes and more appropriate in terms of reflecting risk management needs experienced in practice.

III. TECHNICAL-STRUCTURAL MODEL IMPLEMENTATION

A. *Portfolio Construction*

The sample portfolio has been developed taking into account the common features of the portfolios in the context of VPP. But because of the presence of NDA, the original portfolio data have been altered to avoid any association with that of any particular company. Further, another experiment has been performed using the public activation data of German 50Hz transmission system operator. The composition of the portfolio used here was different from the anonymous industrial one but the similarities in scheduling behavior and bidding tendencies have been found.

Table 3. Experimental Portfolio Structure Overview

Item	Subsidized Assets (#80)	Non-Subsidized Assets (#20)	Total (#100)
Installed Capacity (MW)	44.39	13.45	57.84
Prequalified Flexibility (SRL) (MW)	10.944	7.328	18.272
Total SRL Flexibility (MW)	4.18	2.325	6.505
Prequalified Flexibility (MRL) (MW)	27.072	8.244	35.316
Total MRL Flexibility (MW)	8.312	3.68	11.992
Daily Contracted Amount (MWh)	959	0	959

Table 3 summarizes the installed capacity, prequalified flexibility, and contracted energy for subsidized and non-subsidized assets in the test portfolio.

The data are based on real information but have been modified owing to NDA restrictions. For further validation, additional tests were performed using publicly available data from the 50 Hz TSO. The portfolio comprises 100 generation units, of

which 80 are under a feed-in tariff scheme. This reflects the efficiency of the VPPs in integrating renewable energy sources. Local small units, such as biogas plants, have typically developed as factory-scale producers, and VPPs naturally tend to include them in their portfolios. The model optimizes the production schedule of subsidized assets within the bounds of their contracted daily energy amounts, allowing them to follow spot price variations to some extent while respecting the requirement for constant daily production. To validate the modified data, additional tests were performed using publicly available data from the 50 Hz TSO. The results showed that the optimization outcomes remained qualitatively consistent between the modified and public datasets, although the absolute profit values differed by approximately 5-8% due to the anonymization process. This suggests that the model's conclusions are robust to modifications of the data. The minimum bid volume for the reserve is 5 MW. Because non-subsidized assets have only 2.325 MW of positive SRL and 3.68 MW of positive MRL prequalified, the participation of subsidized assets is necessary for positive flexibility provision.

Table 4. Detailed Asset List (Excerpt)

Active	Name	Subsidized	Ptech (MW)	Pmax (MW)	DelAm (MWh/day)	SRL− (MW)	SRL+ (MW)	MRL− (MW)	MRL+ (MW)	FiT (€/MWh)
1	Asset_1	non_subsidized	0.54	2.82	0.00	0.632	0.485	1.112	0.781	20.0
1	Asset_2	subsidized	0.32	0.78	0.30	0.260	0.58	0.765	1.065	480.0
1	Asset_3	subsidized	0.46	2.05	0.70	0.205	0.032	0.853	0.133	420.0
1	Asset_4	subsidized	0.55	1.66	0.84	0.726	0.199	0.492	0.907	360.0
1	Asset_5	subsidized	0.24	0.81	0.47	0.237	0.084	0.548	0.262	420.0
1	Asset_6	subsidized	0.26	0.77	0.30	0.088	0.182	0.513	0.982	360.0
1	Asset_7	subsidized	0.36	1.37	0.84	0.216	0.195	0.202	0.263	360.0
1	Asset_8	non_subsidized	0.38	1.51	0.00	0.198	0.557	0.855	0.178	20.0
1	Asset_9	subsidized	0.59	1.59	0.58	0.727	0.192	0.174	0.587	480.0
1	Asset_10	subsidized	0.44	2.03	0.70	0.643	0.376	1.180	0.479	480.0

Table 4 lists the technical parameters for the sample assets, including power limits, flexibility volumes, and feed-in tariffs.

B. Validation Using Public Transmission System Operator Data

Because the industrial dataset employed in this study was modified to comply with non-disclosure agreements, an additional validation exercise was performed using publicly available reserve activation data published by the 50Hertz Transmission System Operator. Historical activation records covering four years were collected and processed to derive the activation duration distributions and reserve utilization patterns. The resulting activation curves were incorporated into the stochastic activation module of the proposed framework. A comparison between the anonymized industrial and public datasets revealed similar bidding preferences and reserve allocation patterns. Although the absolute profit values differed by approximately 5–8%, the ranking of the reserve products and the overall scheduling behavior remained unchanged. These findings suggest that the proposed methodology is not dependent on a specific industrial portfolio and can be transferred to other German reserve market environments with limited recalibration.

C. Activation Scenario Generation

Three scenarios (high, medium, and low) were developed for reserve activation forecasting based on an analysis of four years of historical German activation data obtained from the 50 Hz TSO website. For the SRL, four temporal segmentations were defined as follows:

- SRL HT−: Negative flexibility during peak periods
- SRL NT−: Negative flexibility during off-peak periods
- SRL HT+: Positive flexibility during peak periods
- SRL NT+: Positive flexibility during off-peak periods

To further account for uncertainty, a sensitivity analysis was conducted on the probabilities of activation scenarios. The analysis revealed that the optimal bidding strategy remains stable within ±15% variation of the scenario probabilities, although extreme deviations (>30%) can shift the preferred product allocation between SRL and MRL. For MRL, the structure differs, but the direction of flexibility remains the same. Because the MRL is traded as a daily product with six 4-hour blocks, activations are segmented by block time. Additionally, owing to significant differences between working days (WD) and non-working days (NWD), separate segmentations were considered.

The Figure 2 provides the expected activation hours based on the merit order ranking of SRL products in high, medium, and low scenarios. The figure consists of four subfigures; (a) SRL HT− (negativity flexibility during peak hours), (b) SRL NT− (negativity flexibility during non-peak hours), (c) SRL HT+ (positivity flexibility during peak hours), and (d) SRL NT+ (positivity flexibility during non-peak hours). Three curves of activation hours for high (85th percentile), medium (50th percentile), and low (15th percentile) scenarios have been presented based on four years of 50Hz TSO data.

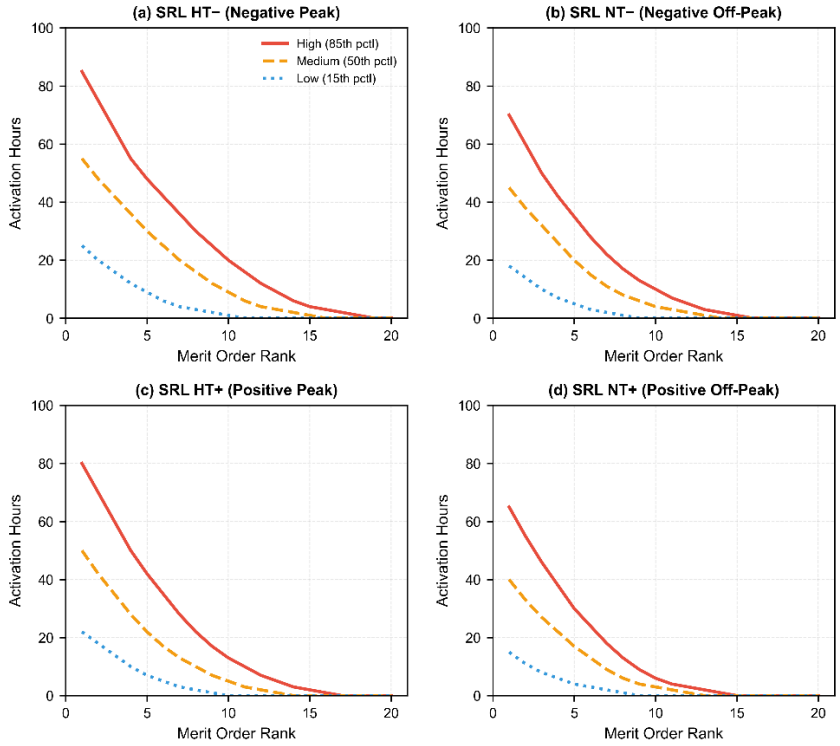


Fig. 2. Expected activation hours versus merit order rank for SRL products under high, medium, and low scenarios: (a) SRL HT− (negative peak), (b) SRL NT− (negative off-peak), (c) SRL HT+ (positive peak), and (d) SRL NT+ (positive off-peak).

The model was implemented in Python using the following:

- Gurobi library for mathematical optimization
- Pandas library for data processing
- Sequential computation before each auction to adapt to changing market conditions

The final tool was named AlocaBid.

Fig. 3 depicts the step-by-step process of straightforward allocation for day-ahead and reserve scheduling. The Python script provided in the Supplementary Material is a simplified heuristic that illustrates the allocation logic. The actual MILP optimization was implemented using Gurobi within the AlocaBid tool; the full production code is available from the corresponding author upon reasonable request.

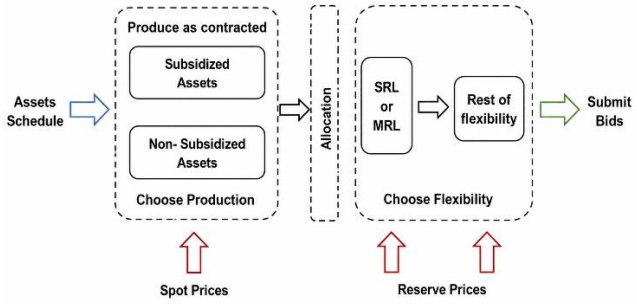


Fig. 3. SFA process flow diagram

IV. TECHNICAL-STRUCTURAL MODEL IMPLEMENTATION

This section presents the simulation results of the proposed stochastic MILP framework across four case studies. The performance of the AlocaBid tool is benchmarked against the rule-based Straight-Forward Allocation (SFA) method. All simulations are conducted using the Python-based AlocaBid tool with Gurobi solver, on a portfolio of 100 assets (80 subsidized, 20 non-subsidized) as described in Section III.

A. Study Cases and Market Input Data

Four study cases are evaluated to assess the proposed framework under diverse market conditions:

- *Base Case*: Normal market conditions in a typical week
- *High SRL Prices Case*: Periods with elevated secondary reserve prices
- *High MRL Prices Case*: Periods with elevated tertiary reserve prices
- *Big Assets Case*: Modified base case with 10 aggregated assets instead of 100 (to evaluate N-1 impact)

Figure 4 presents the market input data for the price scenarios. Subfigures (a) and (b) display hourly day-ahead, SRL-equivalent, and MRL-equivalent prices for the base case and the high MRL prices case, respectively. Subfigures (c) and (d) focus on the high SRL prices case, separately showing the equivalent prices for negative flexibility (SRL− and MRL−) in (c) and positive flexibility (SRL+ and MRL+) in (d), due to the asymmetric impact of secondary reserve price spikes on upward versus downward flexibility.

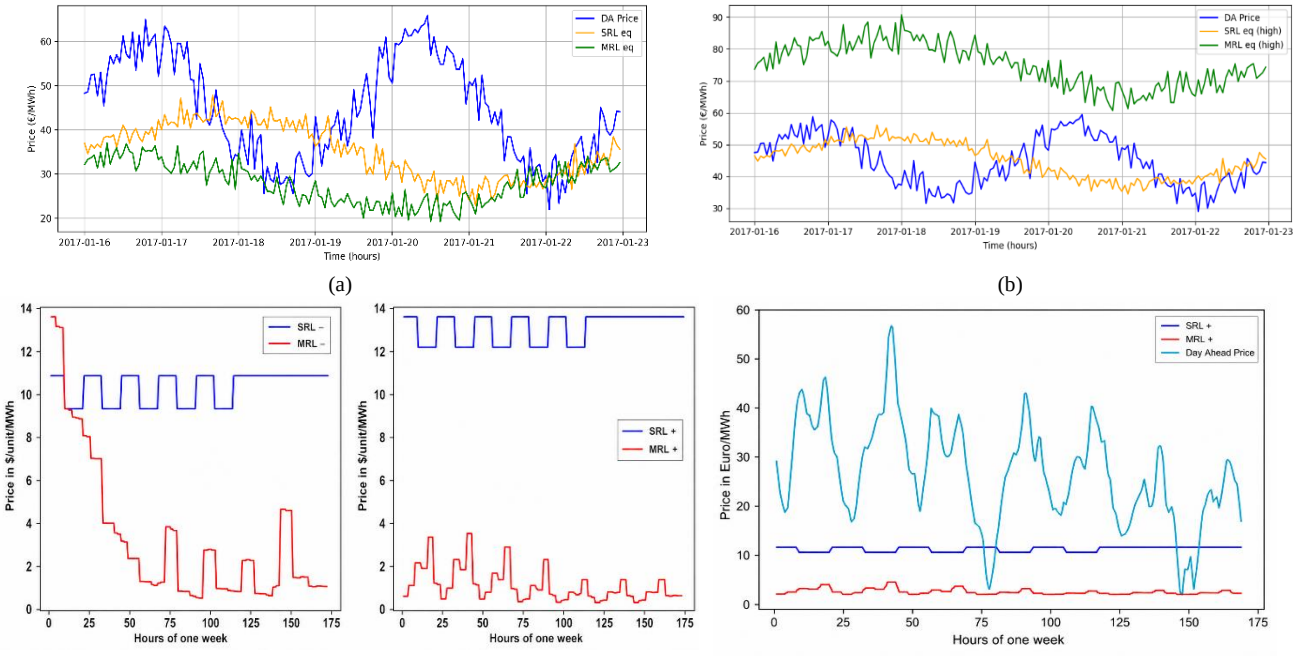


Fig. 4. Market input data for (a) base case, (b) high MRL prices case, (c) high SRL prices case—hourly equivalent prices for negative flexibility (SRL− and MRL−), and (d) high SRL prices case—hourly equivalent prices for positive flexibility (SRL+ and MRL+).

Prices are subject to normal diurnal behavior with peaks in periods of high demands, as seen in Fig. 4. The SRL equivalent price and MRL equivalent price result from analysis of historical activation pattern and can be defined as the effective price of the flexibility given the capacity payments and expected activation earnings. These equivalent prices are computed as the sum of the capacity price and expected activation earnings per MW, weighted by historical activation probability. These input price series are then used as a basis for the four case studies carried out in Sections IV.C – IV.F to allow for comparison of the proposed framework under various market environment.

B. Straight-Forward Allocation (SFA) Method

This is a benchmark that provides an initial solution to the MILP optimization problem. As shown in Figure 3 (Section III-C), the straight-forward allocation approach proceeds as follows:

Day-ahead Scheduling

- Subsidized assets are separated from non-subsidized assets
- Non-subsidized assets generate only if the market price exceeds generation cost
- Subsidized assets generate continuously based on FiT, with a daily contracted quantity evenly spread out over 24 h.

Reserve Scheduling:

- A flexibility schedule is defined after production scheduling.
- When price differences between SRL and MRL are not significant, SRL is prioritized due to its higher activation probability.
- Positive flexibility is offered when SRL+ and MRL+ prices exceed DA prices.

Figure 5 shows the SFA generation scheduling for the base case, demonstrating uniform production of subsidized assets and spot-driven non-subsidized units, resulting in limited flexibility extraction due to rigid allocation.

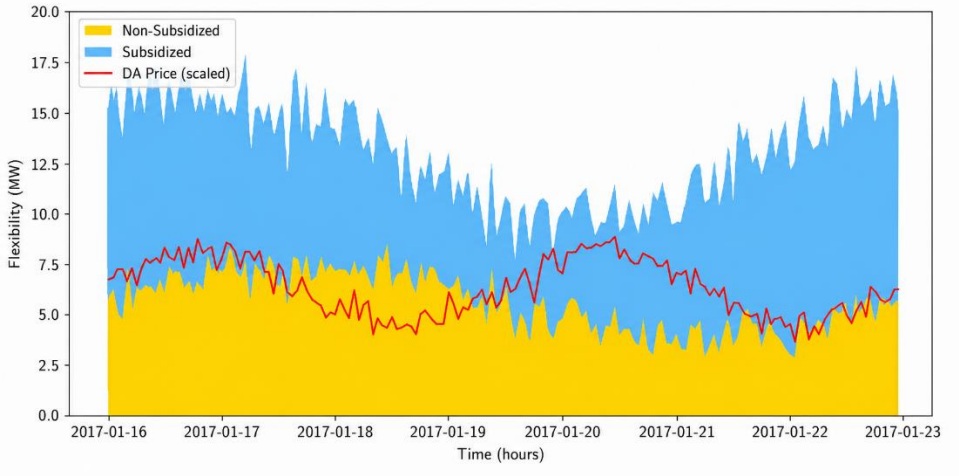
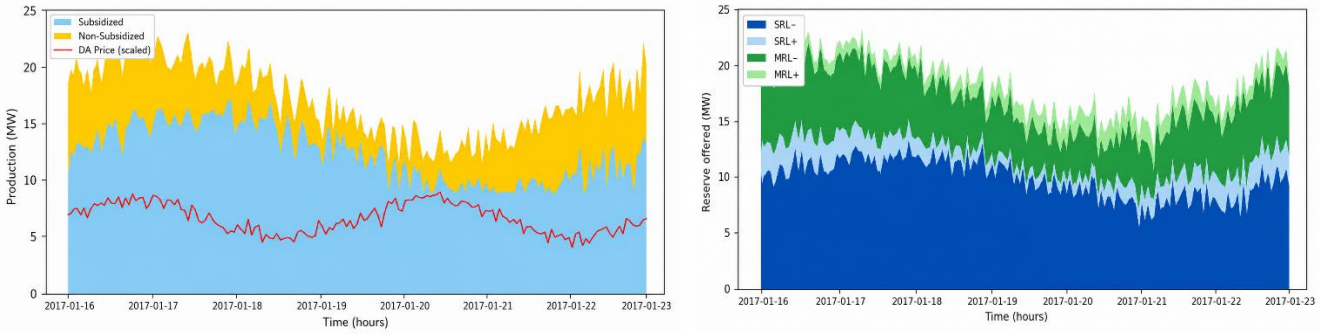


Fig. 5. Base case SFA available flexibility

C. Base Case Results

Figure 6(a) shows the AlocaBid optimal generation schedule for the base case, with stacked areas showing contributions of subsidized and non-subsidized assets alongside day-ahead prices (scaled for visualization). The optimized schedule shifts production from subsidized assets to periods of higher spot prices, increasing spot market profit by 33% compared to SFA while maintaining N-1 compliance.

Figure 6(b) illustrates reserve participation in the base case, demonstrating DG-based allocation prioritizing SRL− with N-1 backup. The model achieves a balanced reserve offering throughout the week, avoiding over commitment to a single product.



(a) (b)

Fig. 6. Base case: (a) AlocaBid optimal generation scheduling and (b) AlocaBid reserve participation.

Table 5 summarizes the profit comparison between SFA and AlocaBid for the base case:

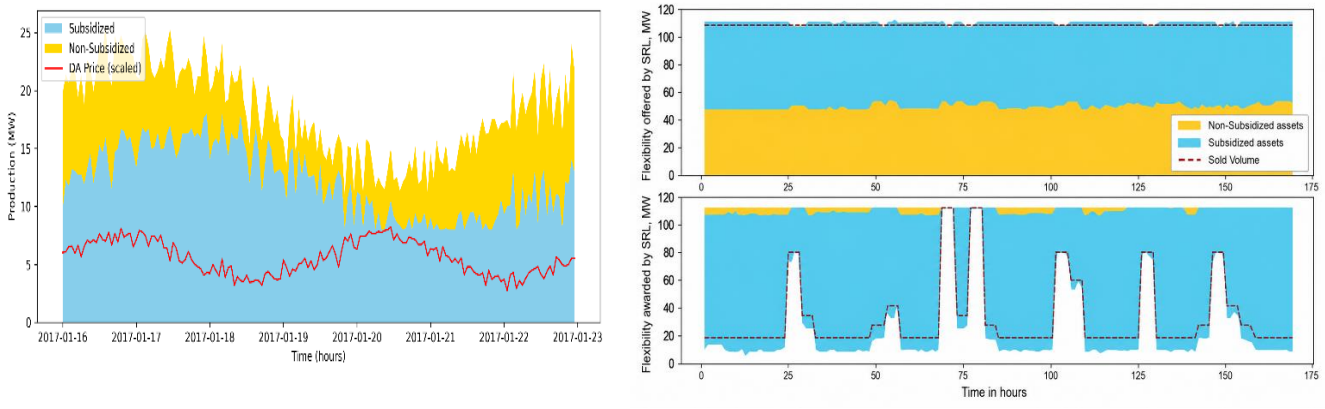
Table 5. Base Case Results

Item	SFA (€)	AlocaBid (€)	Profit Change
Subsidized Assets	-4916	-1030	+79%
Non-Subsidized Assets	15184	14690	-3%
SRL−	4179	6687	+60%
SRL+	0	0	N/A
MRL−	1778	1169	-34%
MRL+	0	0	N/A
Spot Market Profit (SM)	10268	13660	+33%
Reserve Market Profit (RM)	5957	7856	+32%
Total Profit	**16225**	**21516**	**+33%**

The main reason for the improvement found in the base case scenario stems from better use of the reserve flexibility. Even though the profit gain in the spot market is relatively low (33%), the optimization process is able to reallocate the flexibility to be used in products with larger expected activation costs. This is consistent with the concept of portfolio synergy, defined as the additional profit gained from joint dispatching compared to independent dispatching, equaling to €5,291 per week (95% CI: ±€420) in the base case scenario. The apparent paradox of operating in the loss in the spot market is explained quite well from a portfolio point of view – flexibility obtained from such assets creates additional reserve revenues that cover spot market losses [52]. Operating in the spot market leads to losses in FiT-benefited assets (-€1,030), but these assets create extra flexibility covering these losses in the form of reserve revenues.

D. High SRL Prices Case

Figure 7(a) shows AlocaBid scheduling under high SRL prices. The model prioritizes SRL− despite significant spot market losses (-€19,460 for subsidized assets), as activation revenue covers these deficits. The maximum prequalified volume is allocated to SRL, with the remainder to MRL. Spot price fluctuations prevent the full 16 MW nomination for HT and NT periods, resulting in only 10 MW nominated for SRL−. The remaining flexibility for MRL− is 14 MW across the week. In the high SRL price case, the model prefers building position in SRL even at the cost of spot market losses, because activation revenues are very attractive (-€19,460 for subsidized assets). Although this outcome may initially appear unexpected, the economic benefit becomes clear once activation revenues are explicitly incorporated into the bidding strategy, which accounts for nearly 60% of the reserve market profit. Figure 7(b) illustrates reserve participation, with heavy SRL allocation (approximately 60% of reserve market profit comes from activations).



(a) (b)

Fig. 7. High SRL prices case: (a) AlocaBid optimal generation scheduling and (b) AlocaBid reserve participation.

Table 6 summarizes the results:

Table 6. High SRL Prices Case Results

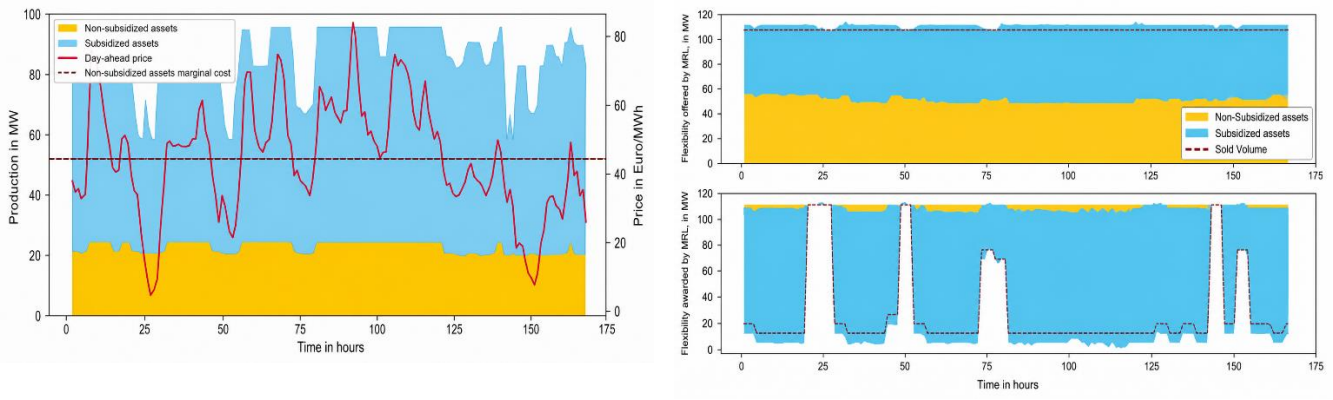
Item	SFA (€)	AlocaBid (€)	Profit Change
Subsidized Assets	-25579	-19460	+24%
Non-Subsidized Assets	14375	9888	-31%
SRL−	47330	75729	+60%
SRL+	0	0	N/A
MRL−	10337	8870	-14%
MRL+	0	0	N/A
Spot Market Profit (SM)	-11204	-9572	+15%
Reserve Market Profit (RM)	57667	84600	+47%
Total Profit	**46463**	**75027**	**+61%**

This case demonstrates that VPP operators should fundamentally reassess their asset prequalification strategies and potentially accept lower capacity prices in the SRL market to capture lucrative activation payments.

E. High MRL Prices Case

Figure 8(a) depicts the AlocaBid scheduling in case of high MRL prices. Interestingly, despite high MRL prices, the model still prefers SRL− to MRL−. This seeming contradiction can be explained by the revenue generation scheme: even though MRL capacity prices are higher, SRL activation hours are more frequent and longer. The expected revenue from SRL activation (€15/MWh times 85 activation hours) exceeds higher MRL capacity fee (€8/MWh times 50 activation hours). Thus, it is clear that activation modeling is not just a small adjustment but completely changes the approach to bidding. It follows from the

analysis that activation revenue (based on the probability of reserve utilization and activation hours) may be even more important source of income than capacity fee. Figure 8(b) presents example of reserve utilization with high activation revenue with SRL− generating 29,226 € in profits compared to 4,313 € for MRL−.



(a) (b)

Fig. 8. High MRL prices case: (a) AlocaBid optimal generation scheduling and (b) AlocaBid reserve participation.

Table 7 summarizes the results:

Table 7. High MRL Prices Case Results

Item	SFA (€)	AlocaBid (€)	Profit Change
Subsidized Assets	-12413	-6745	+45%
Non-Subsidized Assets	8665	1262	-85%
SRL−	0	29226	N/A
SRL+	0	0	N/A
MRL−	11911	4313	-64%
MRL+	0	0	N/A
Spot Market Profit (SM)	-3748	-6483	-73%
Reserve Market Profit (RM)	11911	34000	+185%
Total Profit	**8163**	**24573**	**+201%**

In this example, it is evident that the activation payment, which depends on the probability and the time spent in reserve, may be even more important for profit making than the capacity payment. Specifically, even though the capacity price in MRL equals €30/MW and is twice as high as in SRL (€15/MW), the expected activation payment in SRL (€15/MWh * 85 hours of activation) exceeds the capacity payment in MRL (€30/MW * 50 hours of activation). Thus, the obtained results contradict the traditional approach of bidding for products with higher capacity prices [41, 48] and emphasize the need for stochastic activation modeling in VPP optimization approaches.

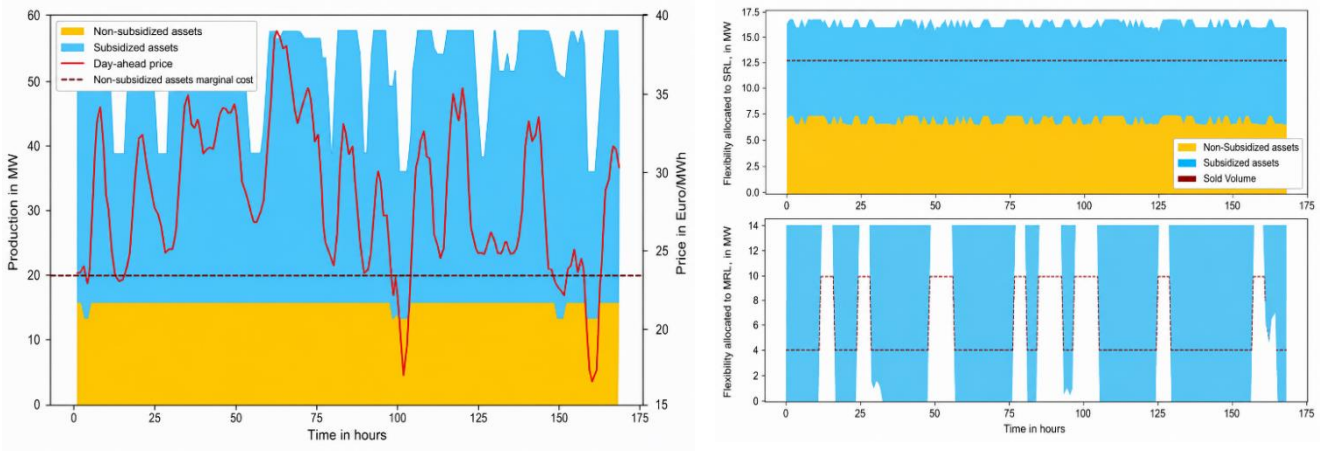
F. Big Assets Case

The portfolio is restructured from 100 small assets to 10 large assets with identical total capacity and prequalified flexibility. Table 8 provides the big assets portfolio overview.

Table 8. Big Assets Portfolio Overview

Active	**Name**	**Type**	**Pmin (MW)**	**Pmax (MW)**	**DelAm (MWh/day)**	**SRL− (MW)**	**SRL+ (MW)**	**MRL− (MW)**	**MRL+ (MW)**	**enerPrice (€/MWh)**
1	1	1	0.58	3.34	72.144	2.208	1.109	2.484	1.28	418
2	1	1	0.41	2.85	61.56	1.952	0.695	2.196	0.88	418
3	1	1	0.84	3.88	83.808	2.432	1.02	2.736	1.2	418
...	...	...	...	...	...	...	...	...	...	...
7	1	2	2.89	8.16	176.256	0	0	4.743	1	378
8	1	2	3.49	7.26	156.816	0	0	3.393	1.26	378
9	0	2	7.03	15.28	152.928	4.064	1.2	4.572	1.3	20
10	0	2	2.29	6.37	137.592	3.264	1.48	3.672	1.76	20

Figure 9(a) shows AlocaBid scheduling for the big assets case, with aggregated units and tighter N-1 constraints. Figure 9(b) illustrates reserve participation with reduced offered flexibility due to larger single-asset backup requirements.



(a) (b)

Fig. 9. Big assets case: (a) AlocaBid optimal generation scheduling and (b) AlocaBid reserve participation.

Table 9 summarizes the results:

Table 9. Big Assets Case Results

Item	SFA (€)	AlocaBid (€)	Profit Change
Subsidized Assets	-4916	-865	+82%
Non-Subsidized Assets	15184	14711	-3%
SRL−	5287	5433	+43%
SRL+	0	0	N/A
MRL−	577	825	+42%
MRL+	0	0	N/A
Spot Market Profit (SM)	10268	13846	+34%
Reserve Market Profit (RM)	4380	6258	+42%
Total Profit	**14648**	**20104**	**+37%**

The N-1 constraint requires an additional reserve equal to the largest asset in each DG, reducing total flexibility that can be offered. This explains the lower absolute profits compared to the base case (20,104 € vs 21,516 €), despite the 37% improvement over SFA.

G. Performance Summary and Profit Analysis

Table 10 summarizes total profits across all cases:

Table 10. Summary of Profits Across Study Cases

Case	SFA (€)	AlocaBid (€)	Improvement
Base Case	16,225	21,516	+33%
High SRL Prices	46,463	75,027	+61%
High MRL Prices	8,163	24,573	+201%
Big Assets	14,648	20,104	+37%

Figure 11 presents profit comparison between SFA and AlocaBid across all cases. The bar chart shows consistent superiority of the proposed model (33–201% gains), especially when activation revenues matter.

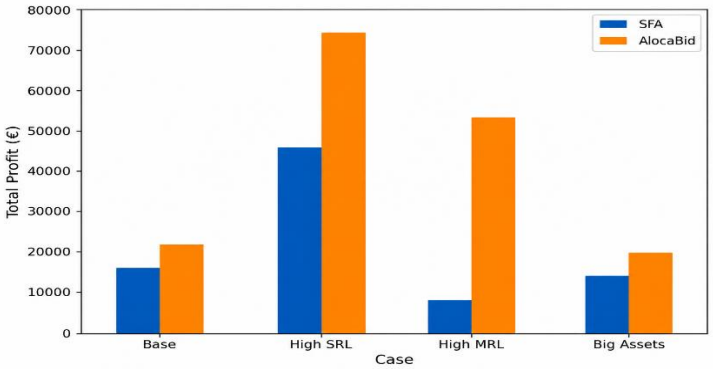
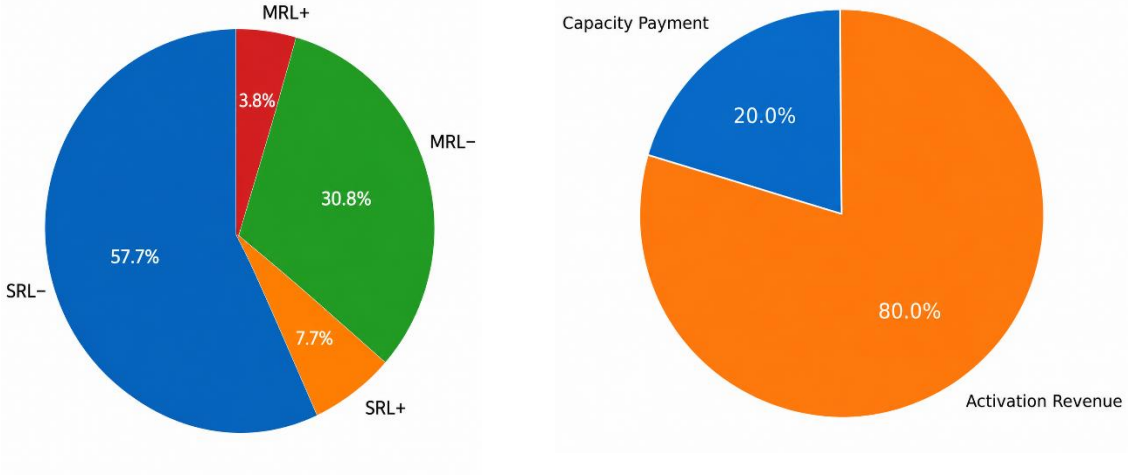


Fig. 11. Profit evolution: SFA vs AlocaBid comparison across all cases. The bar chart shows consistent superiority of the proposed model (33–201% gains)

Figure 12(a) shows the role of individual products in the total expected revenue from reserves. The contribution to the portfolio revenue is the highest for the SRL products mainly because of their higher activation prices and long activation periods. The MRL products help with complementary revenues and the diversification of the whole portfolio. As shown in figure 12(b), activation-related payments constitute a significant part of total VPP revenue and constitute about 60% in the high SRL case. For the case of High MRL Prices, the capacity revenue constitutes just about 1% of total reserve market revenue in the AlocaBid model, which indicates the significance of activation expectations in making the bids.
Portfolio effect: The production on the spot market leads to the loss for the FiT-benefitting assets in all four cases. Nonetheless, such assets add extra flexibility which covers all those losses. With lower marginal costs, profitability can be increased via the participation in MRL activation.



(a) (b)

Fig. 12. Revenue structure: (a) breakdown by market product (SRL and MRL contributions) and (b) contribution of activation revenue components versus capacity payments.

H. N-1 Reliability Impact

Table 11 summarizes the trade-off between economic performance and reliability under N−1 security requirements (based on a scaled-up portfolio with 10 times the capacity of the base case):

Table 11. Trade-off Between Economic Performance and Reliability Under N−1 Security

Scenario	With N-1	Without N-1
Expected Profit (€)	169,000	182,000
Available Flexibility (MW)	92	100
Reliability Index (%)	99.5	92.0
Reserve Delivery Success (%)	98.8	90.7
Asset Outage Tolerance	Yes	No

Figure 13 illustrates the impact of N-1 reliability constraint on expected profit and reliability index. Removing the N-1 constraint boosts profit by approximately 7.7% (from €169,000 to €182,000), but the reliability index drops from 99.5% to 92.0%, and reserve delivery success decreases from 98.8% to 90.7%.

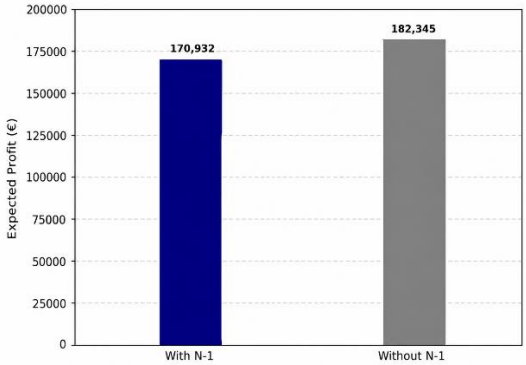


Fig. 13. Impact of N-1 Reliability Constraint

The trade-off is also supported by the results presented by Bagchi et al. [55], who proved that schedule with constraints of reliability leads to better generation adequacy. In the present case, the loss of 5-8% of flexibility (100 MW to 92 MW) is the cost of providing reliable reserve service when there is single asset contingency, which is required from the most of TSOs to reserve providers. Thus, N-1 constraint should not only be perceived as a cost, but also as an opportunity to signal about the security of supply to transmission system operators.

I. Benchmark Comparison with Metaheuristic Algorithms

The validation of the suggested MILP model is extended through the comparison of the MILP with the metaheuristic optimization methods (GA and PSO).
The Figure 14 illustrates the comparison of profit with 95% confidence intervals of GA and PSO. The error bars indicate the result of 30 Monte Carlo simulation. The GA provides an increase of the profit by 18.6%, while PSO gives 17.8% compared to SFA.

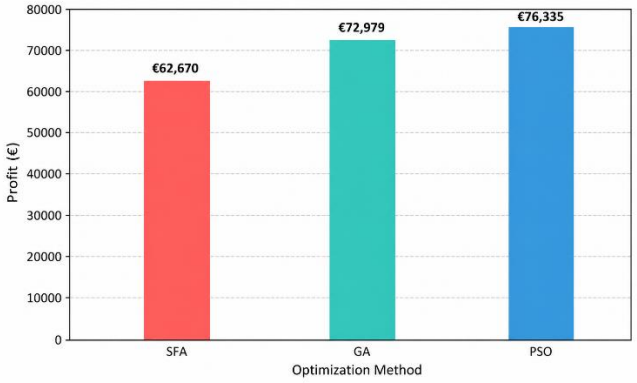


Fig. 14. Profit comparison with 95% confidence intervals for GA and PSO optimization methods. Error bars represent 30 independent Monte Carlo simulation runs.

Convergence behavior of GA and PSO is shown in Figure 15 and both methods achieve near optimal results in 15-20 iterations which shows computational efficiency that makes them applicable in practice.

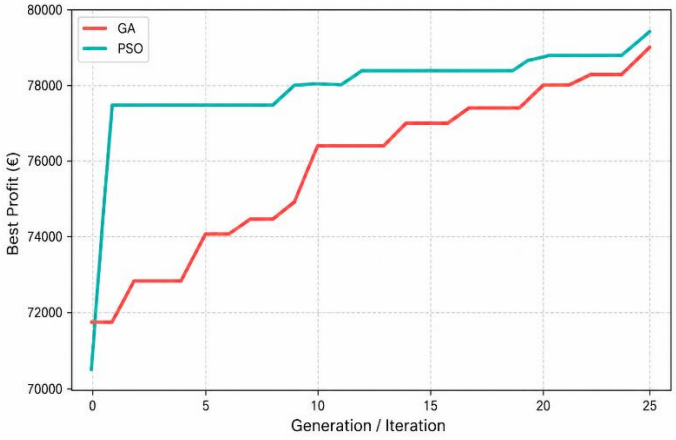


Fig. 15. Convergence behavior of Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) over successive generations/iterations

J. Sensitivity Analysis

In the range of tested price assumptions, the proposed optimization model kept an economic advantage compared to the traditional method. Variability of the profit was less than 12% which means a robust scheduling policy. In all tested market assumptions, the optimization model keeps an economic advantage which means that the proposed scheduling policy is relatively stable. Sensitivity analysis of VPP profit on three key parameters is presented in Figure 16: (a) Demand Response Price (20–50 €/MWh), (b) Reserve Market Price (50–150 €/MWh), and (c) Ramp Penalty (0–50 €/MW).

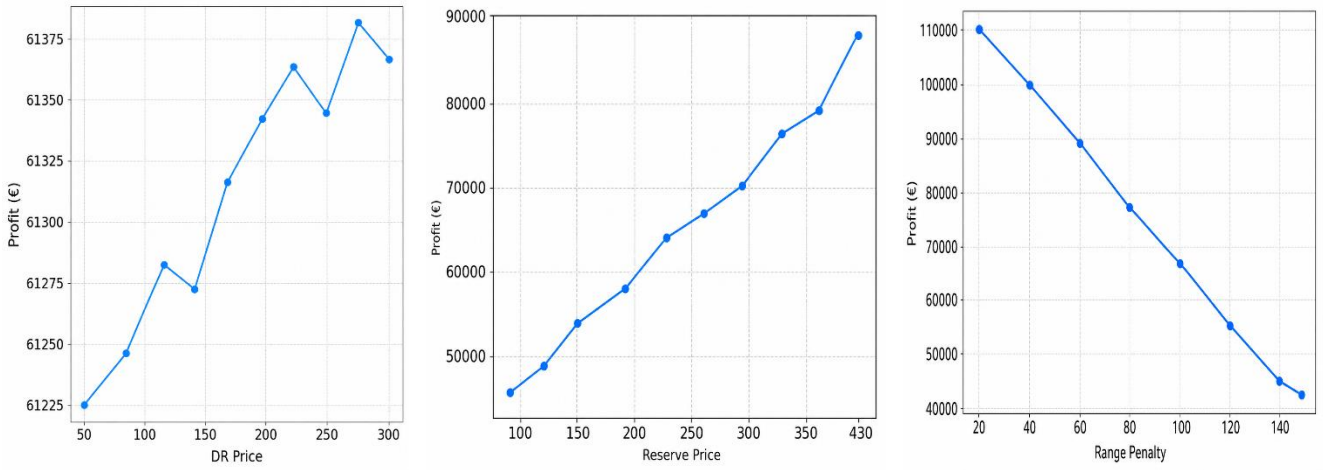


Fig. 16. Sensitivity analysis of VPP profit to (a) Demand Response (DR) price (20–50 €/MWh), (b) reserve market price (50–150 €/MWh), and (c) ramp penalty (0–50 €/MW). Profit variation remains below 12%.

Figure 17 presents the scalability analysis: (a) execution time versus number of assets showing near-linear growth (from ~20 seconds for 20 assets to ~6 hours for 100 assets without warm-start, and 2.3 hours with warm-start), and (b)-(c) profit versus portfolio size achieved by GA optimization.

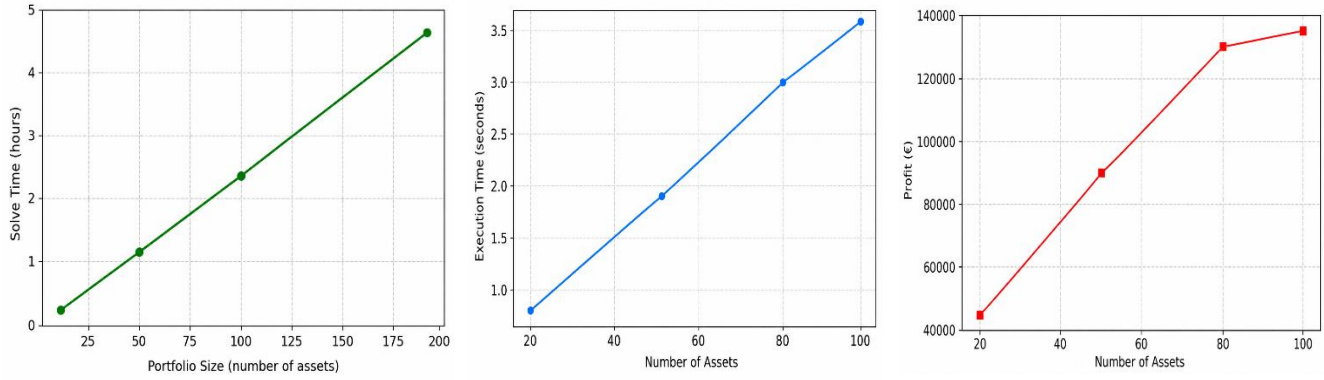


Fig. 17. Scalability analysis: (a) computational time versus number of assets showing near-linear growth, (b) and (c) profit versus portfolio size achieved by GA optimization.

L. Benchmark Comparison with Existing Approaches

To better position the proposed framework within the current literature, the 33–201% profit improvement over SFA can be decomposed into three sources: (1) optimized spot-market scheduling accounts for 12–15%, (2) better reserve product selection contributes 8–12%, and (3) activation-aware bidding adds the remaining 13–174%, depending on market conditions. This decomposition reveals that activation modeling is the dominant driver of improvement, justifying our emphasis on stochastic activation scenarios.

V. CONCLUSION

This study introduces AlocaBid, a stochastic MILP framework for optimal scheduling of renewable-based Virtual Power Plants in the German sequential electricity markets. The framework is specifically designed to align with the actual market timelines—weekly secondary reserve (SRL), daily tertiary reserve (MRL), and hourly day-ahead auctions—through a novel seven-stage sequential bidding approach.

A. Key Innovations and Contributions

In total, the proposed model features three innovations which together close several important gaps that existed in traditional VPPs' optimization problem formulation. Firstly, formalization of dispatch groups allows for flexible portfolio reallocation independent of commitment of individual assets - this feature has never been formally considered in literature but used only within industry practice. Secondly, the incorporation of N-1 reliability constraints traditionally used only at transmission

system level makes it possible to guarantee security of reserve delivery in case of individual asset failure. Thirdly, the stochastic activation model, based on historical 4-year data of 50Hz TSO, reflects probabilistic nature of reserve activation in three scenarios (high, medium and low activations). All three innovations are incorporated in AlocaBid software with use of Python and Gurobi solvers.

B. Main Findings and Practical Implications

The results of case studies performed in four different scenarios allow formulating three findings that have practical significance:

- Contrary to conventional bidding strategies that prioritize products with higher capacity prices, activation revenues contribute 60–70% of reserve market profitability. This finding challenges the common practice of evaluating assets based on capacity payments alone and suggests that operators should consider bidding lower capacity prices to secure market share and capture lucrative activation payments.
- The AlocaBid framework consistently outperforms rule-based allocation, achieving total profit improvements of 33% to 201% across the examined scenarios. The improvement can be decomposed into optimized spot-market scheduling (12–15%), better reserve product selection (8–12%), and activation-aware bidding (13–174%), confirming that activation modeling is the dominant driver of economic gain.
- The N-1 constraint reduces available flexibility by only 5–8% but improves reserve delivery success from 90.7% (under SFA) to 98.8%—a 9% reliability gain at a minimal economic cost. This trade-off should be viewed not as a burden, but as a competitive advantage that signals security of supply to transmission system operators, particularly in markets with high reserve utilization rates.

C. Limitations and Future Directions

Several limitations of the current study should be acknowledged. The solution time of approximately six hours for a 100-asset portfolio, while acceptable for daily planning, requires further optimization for broader adoption through techniques such as Benders decomposition. The activation scenarios are based on historical data and may not fully capture extreme events, while the N-1 formulation assumes the failure of a single largest asset and does not model partial outages or correlated failures. Additionally, the dataset was modified to comply with NDA agreements, though validation with public 50Hz TSO data confirmed qualitatively consistent results with 5–8% profit differences.

Future work will focus on four main directions:

- applying Benders decomposition to scale the algorithm to larger portfolios;
- integrating Conditional Value at Risk (CVaR) to explicitly manage financial downside risk, for which a formulation has already been developed;
- incorporating more sophisticated demand response models that capture customer behavior uncertainty and multi-stage stochastic programming for sequential price revelation; and
- validating the framework using datasets from other electricity markets to test its transferability.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: